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Wearable device

Abstract

A wearable device includes a housing, a main circuit board, two magnetic components and a battery. An inside of the housing has a ring-shaped accommodating space. The main circuit board is mounted in the accommodating space. The main circuit board includes a level section, a first curving section, a second curving section and two electrode contacts. A rear edge of a free side of the first curving section is recessed inward to form a notch. A rear edge of the first curving section is recessed inward to form a first reserving space. A rear edge of the second curving section is recessed inward to form a second reserving space. The two magnetic components are mounted in the accommodating space. The battery is accommodated in the accommodating space. The battery is received in the notch.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is based on, and claims priority from, China Patent Application No. 202320295974.1, filed Feb. 23, 2023, the disclosure of which is hereby incorporated by reference herein in its entirety.

PRIOR ART

(2) Referring to FIG. 7, a conventional wearable device is shown. The conventional wearable device is a ring shape. The wearable device is provided to be worn around a finger of a user for measuring parameters of physical characteristics of the user. The conventional wearable device includes a ring-shaped housing 10', and two magnetic contacts 30' mounted at an inner surface of the ring-shaped housing 10'. When the conventional wearable device is to be charged, the user may make a magnetic charger 50' attracted to the inner surface of the conventional wearable device to charge the conventional wearable device.

(3) Referring to FIG. 7 and FIG. 8, an inside of the conventional wearable device is equipped with a circuit board 20'. The circuit board 20' has two through holes 21'. The two magnetic contacts 30' pass through the two through holes 21'. The two magnetic contacts 30' are soldered to the two through holes 21'. One side of the circuit board 20' defines a notch 22' to accommodate a battery 40'. Because the two magnetic contacts 30' of the conventional wearable device is disposed adjacent to the notch 22', a whole size of the two magnetic contacts 30' is limited by the battery 40', the two magnetic contacts 30' have no way of being disposed to be larger. When the conventional wearable device is charged, a magnetic attraction force between the magnetic charger 50' and the conventional wearable device is weaker, and an enough space is lacked to assemble the larger

magnetic contacts 30' to enhance the magnetic attraction force.

(4) Therefore, it is necessary to provide an innovative wearable device which is easy to adjust a magnetic strength of a magnetic charger.

BACKGROUND OF THE INVENTION

(5) At least one specification heading is required. Please delete this heading section if it is not applicable to your application. For more information regarding the headings of the specification, please see MPEP 608.01(a).

SUMMARY OF THE INVENTION

(6) An object of the present invention is to provide a wearable device which is easy to adjust a magnetic strength of a magnetic charger. The wearable device includes a housing, a main circuit board, two magnetic components and a battery. An inside of the housing has a ring-shaped accommodating space. The main circuit board is bent into a ring shape. The main circuit board is mounted in the accommodating space. The main circuit board includes a level section, a first curving section arched outward from one side of the level section, a second curving section arched outward and opposite to the first curving section from the other side of the level section, and two electrode contacts. A rear edge of a free side of the first curving section is recessed inward to form a notch. A rear edge of the first curving section is recessed inward to form a first reserving space. A rear edge of the second curving section is recessed inward to form a second reserving space. The first reserving space is located between the notch and the second reserving space. The two electrode contacts are slantwise extended downward and towards each other from two free sides of the first curving section and the second curving section. Two free sides of the two electrode contacts are spaced from each other. The two magnetic components are mounted in the accommodating space. Each magnetic component includes a magnet plate accommodated in the accommodating space, and a magnetic contact protruded inward from an inner surface of the magnet plate. The two magnetic contacts of the two magnetic components penetrate through an inner surface of the housing. The two electrode contacts are electrically connected with the two magnet plates of the two magnetic components. The first reserving space and the second reserving space accommodate two rears of the two magnet plates of the two magnetic components. The battery is accommodated in the accommodating space. The battery is received in the notch.

(7) Another object of the present invention is to provide a wearable device. The wearable device is matched with a magnetic charger. The magnetic charger includes a shell, and two magnetic terminals disposed in the shell. The two magnetic terminals are exposed to the shell. The wearable device includes a housing, a main circuit board, two magnetic components and a battery. An inside of the housing has a ring-shaped accommodating space. The main circuit board is bent into a ring shape. The main circuit board is mounted in the accommodating space. The main circuit board includes a level section, a first curving section arched outward from one side of the level section, a second curving section arched outward and opposite to the first curving section from the other side of the level section, and two electrode contacts. A rear edge of a free side of the first curving section is recessed inward to form a notch. A rear edge of the first curving section is recessed inward to form a first reserving space. A rear edge of the second curving section is recessed inward to form a second reserving space. The first reserving space is located between the notch and the second reserving space. The two electrode contacts are slantwise extended downward and towards each other from two free sides of the first curving section and the second curving section. Two free sides of the two electrode contacts are spaced from each other. The two magnetic components are mounted in the accommodating space. Each magnetic component includes a magnet plate accommodated in the accommodating space, and a magnetic contact protruded inward from an inner surface of the magnet plate. The two magnetic contacts of the two magnetic components penetrate through an inner surface of the housing. The two electrode contacts are electrically connected with the two magnet plates of the two magnetic components. The first reserving space and the second reserving space accommodate two rears of the two magnet plates of the two

magnetic components. The battery is accommodated in the accommodating space. The battery is received in the notch. The two magnetic terminals are attracted with the two magnetic contacts of the two magnetic components.

(8) Another object of the present invention is to provide a wearable device matched with a magnetic charger. The magnetic charger includes a shell, and two magnetic terminals disposed in the shell. The two magnetic terminals are exposed to the shell. The wearable device includes a housing, a main circuit board, two magnetic components and a battery. An inside of the housing has a ring-shaped accommodating space. The main circuit board is bent into a ring shape. The main circuit board is mounted in the accommodating space. The main circuit board includes a level section, a first curving section arched outward from one side of the level section, a second curving section arched outward and opposite to the first curving section from the other side of the level section, and two electrode contacts. A rear edge of a free side of the first curving section is recessed inward to form a notch. A rear edge of the first curving section is recessed inward to form a first reserving space. A rear edge of the second curving section is recessed inward to form a second reserving space. The first reserving space is located between the notch and the second reserving space. The two electrode contacts are slantwise extended downward and towards each other from two free sides of the first curving section and the second curving section. Two free sides of the two electrode contacts are spaced from each other. The two magnetic components are mounted in the accommodating space. Each magnetic component includes a magnet plate accommodated in the accommodating space, and a magnetic contact protruded inward from an inner surface of the magnet plate. The two magnetic contacts of the two magnetic components penetrate through an inner surface of the housing. The two electrode contacts are electrically connected with the two magnet plates of the two magnetic components. The first reserving space and the second reserving space accommodate two rears of the two magnet plates of the two magnetic components. The battery is accommodated in the accommodating space. The battery is received in the notch. Widths of the two magnet plates of the two magnetic components are substantially equal to or larger than widths of the two magnetic contacts. The two magnetic terminals contact with the two magnetic components.

(9) As described above, the rear edge of the free side of the first curving section is recessed inward to form the notch for accommodating the battery, the two rear edges of the first curving section and the second curving section are recessed inward to form the first reserving space and the second reserving space for accommodating the two rears of the two magnet plates of the two magnetic components, and the notch is staggered with the first reserving space and the second reserving space to prevent the battery interfering with the two magnet plates of the two magnetic components which are assembled in the first reserving space and the second reserving space. Furthermore, the two magnetic components are set to different widths according to different demands, so that the two magnetic components with the two magnet plates which have the different widths are chosen according to the different demands, so the wearable device is easy to adjust a magnetic strength of the magnetic charger.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The present invention will be apparent to those skilled in the art by reading the following description, with reference to the attached drawings, in which:

(2) FIG. 1 is a perspective view of a wearable device in accordance with a first preferred embodiment of the present invention, wherein the wearable device is matched with a magnetic charger;

(3) FIG. 2 is an exploded view of the wearable device of FIG. 1;

- (4) FIG. 3 is an exploded view of the magnetic charger of FIG. 1;
- (5) FIG. 4 is a perspective view of a circuit board in accordance with the first preferred embodiment of the present invention;
- (6) FIG. 5 is a partially exploded view of the wearable device in accordance with the first preferred embodiment of the present invention;
- (7) FIG. 6 is a partial assembling view of the wearable device in accordance with a second preferred embodiment of the present invention;
- (8) FIG. 7 is a perspective view of a conventional wearable device in prior art, wherein the conventional wearable device is matched with a conventional magnetic charger; and
- (9) FIG. 8 is a perspective view of a circuit board of the conventional wearable device of FIG. 7.

DETAILED DESCRIPTION OF THE INVENTION

- (10) Referring to FIG. 1 and FIG. 2, a wearable device **100** in accordance with a first preferred embodiment of the present invention is shown. The wearable device **100** includes a housing **10**, a main circuit board **20**, two magnetic components **30** and a battery **40**. The wearable device **100** is matched with a magnetic charger **50**.
- (11) The wearable device **100** is provided as an annular shape, and the wearable device **100** is able to be worn around a body surface of a user. An inner surface of the wearable device **100** contacts with the body surface of the user when the wearable device **100** is worn by the user, and an outer surface of the wearable device **100** is opposite to the inner surface of the wearable device **100**. Therefore, in order to describe conveniently, a term “an inward direction of the wearable device **100**” in a following description content of the present invention is defined as a direction that is from the outer surface of the wearable device **100** to the inner surface of the wearable device **100**, and a term “an outward direction of the wearable device **100**” is defined as a direction that is from the inner surface of the wearable device **100** to the outer surface of the wearable device **100**. The outward direction of the wearable device **100** is opposite to the inward direction of the wearable device **100**.
- (12) An inside of the housing **10** has a ring-shaped accommodating space **11**. The main circuit board **20** is bent into a ring shape. The main circuit board **20** is mounted in the accommodating space **11**. Each magnetic component **30** includes a rectangular magnet plate **31** accommodated in the accommodating space **11**, and a magnetic contact **32** protruded inward from an inner surface of the magnet plate **31**. The two magnetic components **30** are mounted to an inner surface of the main circuit board **20**. The main circuit board **20** and the two magnetic components **30** are mounted in the accommodating space **11**. The two magnetic contacts **32** of the two magnetic components **30** penetrate through an inner surface of the housing **10**. The battery **40** is accommodated in the accommodating space **11**.
- (13) Referring to FIG. 1 and FIG. 3, the magnetic charger **50** charges the wearable device **100**. The magnetic charger **50** includes a shell **51**, a charging circuit board **53**, two magnetic terminals **54** and two elastic elements **55**. The two elastic elements **55** are conductive. An inside of the shell **51** has a receiving space **503** penetrating through a first wall **501** of the shell **51**. Two sides of a second wall **502** of the shell **51** have two terminal slots **52** perpendicularly penetrating through the second wall **502** of the shell **51**. The second wall **502** is opposite to the first wall **501**. The charging circuit board **53** is mounted in the receiving space **503** of the inside of the shell **51** for controlling a voltage and a current which are outputted to the wearable device **100**.
- (14) The two magnetic terminals **54** are disposed in the shell **51**. The two magnetic terminals **54** are disposed in the two terminal slots **52**, and the two magnetic terminals **54** are exposed to the second wall **502** of the shell **51**. The two magnetic terminals **54** contact with the two magnetic components **30**, and the two magnetic terminals **54** are combined with the two magnetic components **30**. The two elastic elements **55** are connected between the charging circuit board **53** and the two magnetic terminals **54** to electrically connect the charging circuit board **53** and the two magnetic terminals **54**. The charging circuit board **53** is conductive with the two magnetic terminals **54**.

(15) Referring to FIG. 2, in the first preferred embodiment, in order to assemble the wearable device **100** conveniently, the housing **10** includes an upper housing **12**, a lower housing **13** and a ring bracket **14**. The upper housing **12** is an arc shape. The lower housing **13** is a semicircle shape. The upper housing **12** is covered to an upper portion of the ring bracket **14**, and the lower housing **13** is covered to a lower portion of the ring bracket **14**. The upper housing **12** is combined with the lower housing **13** to form a circle shape. The ring bracket **14** is provided with an inner ring wall **14a** and two radial walls **14d**. The upper housing **12** is spaced from the inner ring wall **14a** of the ring bracket **14**. The lower housing **13** is spaced from the inner ring wall **14a** of the ring bracket **14**. The upper housing **12**, the lower housing **13** and the ring bracket **14** surround the accommodating space **11** in the housing **10**.

(16) The inner ring wall **14a** has a horizontal section **14b** arranged at a top of the inner ring wall **14a**, and a semicircular section **14c** arched outward and towards each other from two sides of the horizontal section **14b**. The two radial walls **14d** are bent outward and are extended radially from two end peripheries of the inner ring wall **14a**. A bottom of the inner ring wall **14a** defines two through holes **15** penetrating through the bottom of the inner ring wall **14a**. Several portions of a periphery of a top surface of the horizontal section **14b** protrude upward to form a plurality of positioning bumps **16**.

(17) Referring to FIG. 2, FIG. 4 and FIG. 5, the main circuit board **20** includes a level section **21**, a first curving section **22**, a second curving section **23** and two electrode contacts **24**. The first curving section **22** is arched outward from one side of the level section **21**. The second curving section **23** is arched outward and opposite to the first curving section **22** from the other side of the level section **21**. Two free sides of the first curving section **22** and the second curving section **23** are disconnected from each other. The two electrode contacts **24** are slantwise extended downward and towards each other from the two free sides of the first curving section **22** and the second curving section **23**. Two free sides of the two electrode contacts **24** are spaced from each other.

(18) The two magnetic components **30** are positioned at two inner surfaces of the two electrode contacts **24**. The two magnet plates **31** of the two magnetic components **30** contact with the two inner surfaces of the two electrode contacts **24**. The two electrode contacts **24** are electrically connected with the two magnet plates **31** of the two magnetic components **30**. A rear edge of the free side of the first curving section **22** is recessed inward to form a notch **25** for accommodating the battery **40**. The battery **40** is received in the notch **25**. A rear edge of the first curving section **22** is recessed inward to form a first reserving space **26**. A rear edge of the second curving section **23** is recessed inward to form a second reserving space **27**. The first reserving space **26** and the second reserving space **27** accommodate two rears of the two magnet plates **31** of the two magnetic components **30**. The first reserving space **26** is located between the notch **25** and the second reserving space **27**. The notch **25** is staggered with the first reserving space **26** and the second reserving space **27** to prevent the battery **40** interfering with the two magnet plates **31** of the two magnetic components **30** which are assembled in the first reserving space **26** and the second reserving space **27**. Several portions of a front edge and a rear edge of the level section **21** are recessed inward to form a plurality of positioning holes **28** penetrating through a top surface and a bottom surface of the level section **21**. The plurality of the positioning holes **28** are arranged corresponding to the plurality of the positioning bumps **16**.

(19) Referring to FIG. 1 to FIG. 6, when the wearable device **100** is assembled, the plurality of the positioning holes **28** are aligned with the plurality of the positioning bumps **16**, and the plurality of the positioning bumps **16** are fixed in the plurality of the positioning holes **28**. Then the level section **21** is horizontally placed on the horizontal section **14b** to fasten the main circuit board **20** to the ring bracket **14**. The level section **21**, the first curving section **22** and the second curving section **23** are accommodated into the accommodating space **11**. The battery **40** is inserted into the notch **25**, and the battery **40** is electrically connected with the main circuit board **20**. The two magnetic contacts **32** penetrate through the two through holes **15**. The two magnet plates **31** of the two

magnetic components **30** are adhered with the two electrode contacts **24** by a conductive adhesive to make that the two magnetic components **30** are electrically connected with the main circuit board **20**.

(20) Referring to FIG. **1** and FIG. **3**, when the user wants to charge the wearable device **100**, the user places the magnetic charger **50** inside the wearable device **100**, the two magnetic terminals **54** are attracted with the two magnetic contacts **32** of the two magnetic components **30**, and the two magnetic terminals **54** are aligned with the two magnetic contacts **32** of the two magnetic components **30** to make that the magnetic charger **50** is magnetically attracted to the wearable device **100**, and an electrical conduction is formed between the wearable device **100** and the magnetic charger **50**.

(21) Referring to FIG. **1**, FIG. **5** and FIG. **6**, the wearable device **100** in accordance with a second preferred embodiment of the present invention is shown. Differences between the wearable device **100** in accordance with the first preferred embodiment of the present invention and the wearable device **100** in accordance with the second preferred embodiment of the present invention are described as follows. In the first preferred embodiment, widths of the two magnet plates **31** of the two magnetic components **30** are substantially equal to widths of the two magnetic contacts **32**. In the second preferred embodiment, the widths of the two magnet plates **31** of the two magnetic components **30** are larger than the widths of the two magnetic contacts **32**. Partial portions of the two magnet plates **31** of the two magnetic components **30** are received in the first reserving space **26** and the second reserving space **27**, so the two magnet plates **31** of the two magnetic components **30** have larger contact areas, and a greater attraction force is generated between the two magnet plates **31** of the two magnetic components **30** and the two magnetic terminals **54** of the magnetic charger **50**.

(22) Referring to FIG. **1** to FIG. **6**, in the present invention, the widths of the two magnet plates **31** of the two magnetic components **30** are adjustable, the widths of the two magnet plates **31** of the two magnetic components **30** are substantially equal to or larger than the widths of the two magnetic contacts **32**, so a magnetic strength between the magnetic charger **50** and the wearable device **100** is easily adjusted.

(23) As described above, the rear edge of the free side of the first curving section **22** is recessed inward to form the notch **25** for accommodating the battery **40**, the two rear edges of the first curving section **22** and the second curving section **23** are recessed inward to form the first reserving space **26** and the second reserving space **27** for accommodating the two rears of the two magnet plates **31** of the two magnetic components **30**, and the notch **25** is staggered with the first reserving space **26** and the second reserving space **27** to prevent the battery **40** interfering with the two magnet plates **31** of the two magnetic components **30** which are assembled in the first reserving space **26** and the second reserving space **27**. Furthermore, the two magnetic components **30** are set to different widths according to different demands, so that the two magnetic components **30** with the two magnet plates **31** which have the different widths are chosen according to the different demands, so the wearable device **100** is easy to adjust the magnetic strength of the magnetic charger **50**.

Claims

1. A wearable device, comprising: a housing, an inside of the housing having a ring-shaped accommodating space; a main circuit board bent into a ring shape, the main circuit board being mounted in the accommodating space, the main circuit board including: a level section; a first curving section arched outward from one side of the level section, a rear edge of a free side of the first curving section being recessed inward to form a notch, a rear edge of the first curving section being recessed inward to form a first reserving space; a second curving section arched outward and opposite to the first curving section from the other side of the level section, a rear edge of the

second curving section being recessed inward to form a second reserving space, the first reserving space being located between the notch and the second reserving space; and two electrode contacts slantwise extended downward and towards each other from two free sides of the first curving section and the second curving section, two free sides of the two electrode contacts being spaced from each other; two magnetic components mounted in the accommodating space, each magnetic component including a magnet plate accommodated in the accommodating space, and a magnetic contact protruded inward from an inner surface of the magnet plate, the two magnetic contacts of the two magnetic components penetrating through an inner surface of the housing, the two electrode contacts being electrically connected with the two magnet plates of the two magnetic components, the first reserving space and the second reserving space accommodating two rears of the two magnet plates of the two magnetic components; and a battery accommodated in the accommodating space, the battery being received in the notch.

2. The wearable device as claimed in claim 1, wherein the housing includes an upper housing, a lower housing and a ring bracket, the upper housing is an arc shape, the lower housing is a semicircle shape, the upper housing is covered to an upper portion of the ring bracket, and the lower housing is covered to a lower portion of the ring bracket, the upper housing is combined with the lower housing to form a circle shape, the upper housing, the lower housing and the ring bracket surround the accommodating space in the housing.

3. The wearable device as claimed in claim 2, wherein the ring bracket being provided with an inner ring wall and two radial walls, the inner ring wall has a horizontal section arranged at a top of the inner ring wall, and a semicircular section arched outward and towards each other from two sides of the horizontal section, the two radial walls are bent outward and are extended radially from two end peripheries of the inner ring wall, a bottom of the inner ring wall defines two through holes penetrating through the bottom of the inner ring wall, the two magnetic contacts penetrate through the two through holes.

4. The wearable device as claimed in claim 3, wherein several portions of a periphery of a top surface of the horizontal section protrude upward to form a plurality of positioning bumps, several portions of a front edge and a rear edge of the level section are recessed inward to form a plurality of positioning holes penetrating through a top surface and a bottom surface of the level section, the plurality of the positioning holes are arranged corresponding to the plurality of the positioning bumps, the plurality of the positioning bumps are fixed in the plurality of the positioning holes.

5. The wearable device as claimed in claim 1, wherein widths of the two magnet plates of the two magnetic components are substantially equal to widths of the two magnetic contacts.

6. The wearable device as claimed in claim 1, wherein widths of the two magnet plates of the two magnetic components are larger than widths of the two magnetic contacts, partial portions of the two magnet plates of the two magnetic components are received in the first reserving space and the second reserving space.

7. The wearable device as claimed in claim 1, wherein the notch is staggered with the first reserving space and the second reserving space.

8. A wearable device matched with a magnetic charger, the magnetic charger including a shell, and two magnetic terminals disposed in the shell, the two magnetic terminals being exposed to the shell, the wearable device comprising: a housing, an inside of the housing having a ring-shaped accommodating space; a main circuit board bent into a ring shape, the main circuit board being mounted in the accommodating space, the main circuit board including: a level section; a first curving section arched outward from one side of the level section, a rear edge of a free side of the first curving section being recessed inward to form a notch, a rear edge of the first curving section being recessed inward to form a first reserving space; a second curving section arched outward and opposite to the first curving section from the other side of the level section, a rear edge of the second curving section being recessed inward to form a second reserving space, the first reserving space being located between the notch and the second reserving space; and two electrode contacts

slantwise extended downward and towards each other from two free sides of the first curving section and the second curving section, two free sides of the two electrode contacts being spaced from each other; two magnetic components mounted in the accommodating space, each magnetic component including a magnet plate accommodated in the accommodating space, and a magnetic contact protruded inward from an inner surface of the magnet plate, the two magnetic contacts of the two magnetic components penetrating through an inner surface of the housing, the two electrode contacts being electrically connected with the two magnet plates of the two magnetic components, the first reserving space and the second reserving space accommodating two rears of the two magnet plates of the two magnetic components; and a battery accommodated in the accommodating space, the battery being received in the notch; wherein the two magnetic terminals are attracted with the two magnetic contacts of the two magnetic components.

9. A wearable device matched with a magnetic charger, the magnetic charger including a shell, and two magnetic terminals disposed in the shell, the two magnetic terminals being exposed to the shell, the wearable device comprising: a housing, an inside of the housing having a ring-shaped accommodating space; a main circuit board bent into a ring shape, the main circuit board being mounted in the accommodating space, the main circuit board including: a level section; a first curving section arched outward from one side of the level section, a rear edge of a free side of the first curving section being recessed inward to form a notch, a rear edge of the first curving section being recessed inward to form a first reserving space; a second curving section arched outward and opposite to the first curving section from the other side of the level section, a rear edge of the second curving section being recessed inward to form a second reserving space, the first reserving space being located between the notch and the second reserving space; and two electrode contacts slantwise extended downward and towards each other from two free sides of the first curving section and the second curving section, two free sides of the two electrode contacts being spaced from each other; two magnetic components mounted in the accommodating space, each magnetic component including a magnet plate accommodated in the accommodating space, and a magnetic contact protruded inward from an inner surface of the magnet plate, the two magnetic contacts of the two magnetic components penetrating through an inner surface of the housing, the two electrode contacts being electrically connected with the two magnet plates of the two magnetic components, the first reserving space and the second reserving space accommodating two rears of the two magnet plates of the two magnetic components; and a battery accommodated in the accommodating space, the battery being received in the notch; wherein widths of the two magnet plates of the two magnetic components are substantially equal to or larger than widths of the two magnetic contacts, the two magnetic terminals contact with the two magnetic components.
